Geometric Limits of Defense Design: Impossibility Theorems for Prompt Injection Defenses

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Abstract

Prior work on LLM safety has focused on discovering adversarial inputs or bounding perturbation robustness around individual examples. We ask a different question: what are the *geometric limits of defense design itself*?

We prove impossibility results via two independent paths—continuous and discrete—so that neither relies on the other.

Path A (Continuous). Three results of increasing strength: (1) any continuous, utility-preserving defense on a connected Hausdorff space must fix boundary points; (2) when both defense and alignment function are Lipschitz, the defense cannot uniformly reduce alignment deviation far below threshold on a positive-measure band; (3) under a transversality condition, the defense leaves a positive-measure region genuinely unsafe: $f(D(x)) > \tau$ persists over volume.

Path B (Discrete). On finite ordered sets with no topology: (1) boundary crossings exist by a discrete IVT; (2) utility-preserving defenses must leave the safe endpoint of each crossing fixed; (3) any defense with bounded capacity is overwhelmed by the exponentially growing attack surface.

The Tietze extension theorem bridges the two paths, but Path B stands alone for reviewers who object to continuous relaxation. The full theory—36 files, ∼300 theorems, zero `sorry`, three standard axioms—is machine-verified in Lean 4 with Mathlib and validated against three frontier LLMs.

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities?

The defense mechanisms people have proposed—input classifiers [1], output filters [9], constitutional rewriting [3]—all share the same structure: a function $D\colon X \to X$ that preprocesses prompts, aiming to map unsafe inputs to safe equivalents while leaving safe inputs alone.

Prior work has focused on discovering or mitigating individual adversarial inputs. We ask a different question: what are the *geometric limits of the defense mapping itself*? We show that under mild design constraints, no defense can eliminate the safety boundary. Vulnerabilities arise not from specific inputs but from the structure of the space and the constraints on admissible defenses.

We prove this via two independent paths, so that neither relies on the other.

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Path A: Continuous impossibility (three tiers). *Tier 1 (Boundary fixation).* If D is continuous, utility-preserving, and the space is connected, D must fix boundary points (Theorem 4.1).

Tier 2 (ε-robust constraint). If D is K-Lipschitz and f is L-Lipschitz:

$$f(D(x)) \;\geq\; \tau - L(K+1)\,\delta \tag{1}$$

for x within distance δ of a fixed boundary point (Theorem 5.1). The ε-band has positive measure (Theorem 5.2).

Tier 3 (Persistent unsafe region). Under transversality (boundary slope $> L(K+1)$), a positive-measure region satisfies $f(D(x)) > \tau$—genuinely unsafe after defense (Theorem 6.2).

Path B: Discrete impossibility (no topology). *Discrete IVT.* On $\{0, \ldots, n\}$, if $f(0) < \tau \leq f(n)$, consecutive indices cross τ (Theorem 7.1).

Discrete defense fixation. If D fixes safe vertices, the safe side of each crossing is fixed (Theorem 7.2).

Capacity exhaustion. Any defense with $< N$ distinct behaviors misclassifies at least one of N adversarial configurations (Theorem 7.3). The attack surface grows as 2^d, overwhelming any fixed budget (Theorem 7.4).

Bridging the paths. The Tietze extension theorem (Theorem 8.1) shows that discrete data admits continuous extensions for which Path A holds. But Path B stands alone: it requires no continuous relaxation, no Tietze, no topology.

Why continuity? Continuity captures a natural requirement: small input changes should not produce arbitrarily large output changes. Any defense based on learned representations or differentiable processing satisfies this. Discontinuous defenses (blocklists, keyword filters) escape Path A but not Path B.

Contributions.

- **Path A**: Three-tier continuous impossibility (Sections 4 to 6).
- **Path B**: Discrete impossibility without topology (Section 7).
- Tietze bridge connecting the two (Section 8).
- Counterexamples proving each hypothesis necessary (Section 4.1).
- Supporting theory: basins, convergence, transferability, authority, gradient, stability, cost (Sections 9 to 12).
- Empirical validation on three LLMs (Section 13).
- Lean 4 artifact: 36 files, ∼300 theorems, zero `sorry` (Section 15).

2 Related Work

Adversarial robustness. Unlike adversarial robustness results [14, 8, 7], which bound perturbation radii around individual inputs, our result is a *structural constraint on the defense mapping itself*, independent of perturbation models, norms, or training procedures. Certified defenses [13, 10] guarantee local neighborhoods; we constrain *global* defense behavior.

Prompt injection. Attack methods [17, 6, 11, 12] discover individual failures. The Manifold of Failure framework [4] maps their topology. Our result shows these failures are not incidental but *structurally unavoidable.*

Impossibility results. No-free-lunch theorems [16] limit average performance across tasks. Robustness–accuracy tradeoffs [15] are distributional. Our result holds *pointwise* for any continuous defense on a *fixed task*, and the discrete path holds without any continuity assumption at all.

Topological views. While our continuous proof uses elementary topology, the novelty is identifying that *standard design constraints induce a forced fixed-point structure*. The persistent unsafe region (Tier 3) and the discrete path go beyond topology entirely.

Formal verification. To our knowledge, this is the first formally verified impossibility result for prompt injection defense [2, 5].

3 Setup

Definition 3.1 (Alignment Deviation). *A continuous map $f\colon X \to \mathbb{R}$ with threshold τ partitions X into $S_\tau = \{f < \tau\}$ (safe), $U_\tau = \{f > \tau\}$ (unsafe), and $B_\tau = \{f = \tau\}$ (boundary).*

Definition 3.2 (Defense). *A continuous map $D\colon X \to X$ is* utility-preserving *if $D|_{S_\tau} = \mathrm{id}$, and* complete *if $f(D(x)) < \tau$ for all x.*

4 Path A, Tier 1: Boundary Fixation

Theorem 4.1 (Boundary Fixation). *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ continuous with $D|_{S_\tau} = \mathrm{id}$. Then $\exists\, z$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies this. This set is nonempty.*

Proof. (1) $\mathrm{Fix}(D)$ is closed (Hausdorff $\Rightarrow$ diagonal closed). (2) $S_\tau \subseteq \mathrm{Fix}(D)$; taking closures, $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. (3) S_τ is open and not closed (else clopen in connected space, contradiction). (4) $\overline{S_\tau} \supsetneq S_\tau$; pick $z \in \overline{S_\tau} \setminus S_\tau$: $f(z) = \tau$. (5) $z \in \mathrm{Fix}(D)$, so $D(z) = z$. □

Corollary 4.2. *No continuous utility-preserving defense is complete.*

4.1 Each Hypothesis Is Necessary

Counterexample 4.3 (Disconnected). *$X = \{0, 1\}$ discrete: $D(1) = 0$ is complete.*

Counterexample 4.4 (Discontinuous). *$D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: complete.*

Counterexample 4.5 (Not utility-preserving). *$D(x) = x_0$ (constant): complete.*

Remark 4.6 (Defense trilemma). *At most two of: continuity, utility preservation, completeness.*

5 Path A, Tier 2: ε-Robust Constraint

Theorem 5.1 (ε-Robust Constraint). *Under Theorem 4.1's hypotheses, if f is L-Lipschitz and D is K-Lipschitz, the fixed point z satisfies: for all x (strongest near z), $f(D(x)) \geq \tau - L(K+1)\,\mathrm{dist}(x, z)$.*

Theorem 5.2 (Positive-Measure Band). *The ε-band $\{\tau - \varepsilon \leq f \leq \tau\}$ has positive measure (under any measure positive on nonempty open sets). It contains $\mathrm{ball}(c, \varepsilon/(4L))$ for c with $f(c) = \tau - \varepsilon/2$.*

Remark 5.3 (Depth, not direction). *Tier 2 constrains how* far below *τ the defense can push near-boundary points. It does not prove they remain unsafe—the defense may map them to $[\tau - L(K + 1)\delta,\ \tau)$.*

6 Path A, Tier 3: Persistent Unsafe Region

Definition 6.1 (Steep region). $\mathcal{S} = \{x : f(x) > \tau + L(K+1)\,\mathrm{dist}(x,z)\}$*—points where alignment deviation exceeds* τ *by more than the defense can compensate.*

Theorem 6.2 (Persistent Unsafe Region). *If* $\mathcal{S} \neq \emptyset$*: (1)* $\mathcal{S}$ *is open; (2)* $\mu(\mathcal{S}) > 0$*; (3)* $f(D(x)) > \tau$ *for all* $x \in \mathcal{S}$*.*

Proof. (1) $x \mapsto f(x) - L(K+1)\,\mathrm{dist}(x,z)$ is continuous; $\mathcal{S}$ is its superlevel set at τ. (2) Open + nonempty $\Rightarrow$ positive measure. (3) $f(D(x)) \geq f(x) - L(K+1)\,\mathrm{dist}(x,z) > \tau$ for $x \in \mathcal{S}$. □

Remark 6.3 (Transversality). $\mathcal{S} \neq \emptyset$ *whenever the boundary has directional slope* $> L(K+1)$*—the surface rises faster than the defense can pull it down. Verified in Lean: if* f *has directional derivative* $c > L(K+1)$ *along unit vector* v *at* z*, then* $z + tv \in \mathcal{S}$ *for small* $t > 0$*.*

7 Path B: Discrete Impossibility

Path B requires no topology, no continuity, no Tietze extension.

Theorem 7.1 (Discrete IVT). *Let* $f\colon \{0,\ldots,n{+}1\} \to \mathbb{R}$ *with* $f(0) < \tau$ *and* $f(n{+}1) \geq \tau$*. Then* $\exists\, i$ *with* $f(i) < \tau$ *and* $f(i{+}1) \geq \tau$*.*

Proof. By induction: if $f(0) < \tau$ and no consecutive crossing exists, then $f(k) < \tau$ for all k, contradicting $f(n+1) \geq \tau$. □

Theorem 7.2 (Discrete Defense Fixation). *If* D *fixes safe vertices (*$f(v) < \tau \Rightarrow D(v) = v$*), the crossing from Theorem 7.1 has its safe endpoint fixed.*

Theorem 7.3 (Capacity Exhaustion). *If there are* m *adversarial configurations and the defense realizes* $< m$ *distinct behaviors, at least two configurations receive the same treatment (pigeonhole).*

Theorem 7.4 (Exponential Attack Surface). $2^d \geq d+1$ *for all* d*. For any fixed budget* B*, there exists* d_0 *with* $2^d > B$ *for all* $d \geq d_0$*. The attack surface grows exponentially; no fixed-capacity defense suffices.*

Remark 7.5 (Discrete trilemma). *On a finite ordered set: if* f *crosses* τ *and* D *fixes safe inputs,* D *must leave the safe endpoint of the crossing fixed. This is the discrete boundary fixation, proved by induction rather than topology.*

8 Bridge: Tietze Extension

Theorem 8.1 (Continuous Relaxation). *Let* X *be connected, normal, Hausdorff with finite observations* $g(p) < \tau < g(q)$*. Then* $\exists$ *continuous* f *agreeing with all observations for which Path A holds. Lipschitz extensions enable Tiers 2–3.*

Remark 8.2. *Path B does not need Theorem 8.1. The bridge connects the two paths but neither depends on the other for its core claims.*

9 Vulnerability Landscape

Theorem 9.1 (Basin Structure). *U_τ is open; positive measure under any measure positive on nonempty open sets.*

Theorem 9.2 (Fragment Size). *L-Lipschitz f with $f(p) > \tau$: connected component of U_τ at p has diameter $\geq 2(f(p) - \tau)/L$.*

10 Attack Properties

Theorem 10.1 (Perturbation Robustness). $\mathrm{ball}(p, (f(p) - \tau)/L) \subseteq U_\tau$*; radius monotone in $f(p)$.*

Theorem 10.2 (Convergence). *Monotone-improvement operators on $[0, 1]$ converge in $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem 10.3 (Transferability). $\|f - g\|_\infty \leq \delta \Rightarrow \{f > \tau + \delta\} \subseteq \{g > \tau\}$*. Zero-cost transfer.*

Theorem 10.4 (Authority Monotonicity). *$f(y, \cdot)$ monotone in authority $\Rightarrow$ vulnerability upward-closed; critical threshold exists by IVT when $f(y, \cdot)$ is continuous and crosses τ. If also monotone in indirection, threshold curve is non-increasing.*

Theorem 10.5 (Gradient Ascent). *Nonzero Fréchet derivative at $x \Rightarrow \exists\, v, \varepsilon > 0$: $f(x+\varepsilon v) > f(x)$.*

11 Stability Under Fine-Tuning

Theorem 11.1 (Interior Stability). $\|f - g\|_\infty \leq \varepsilon$*: deep basin points and deep safe points preserve classification. Only $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem 11.2 (Crossing Preservation). *Transversal crossings persist under ε-perturbation.*

Theorem 11.3 (Patching Is Nonlocal). *Eliminating a vulnerability at one point necessarily changes values elsewhere. (Explicit Lean construction.)*

12 Cost Asymmetry

Theorem 12.1 (Grid-Based Cost Asymmetry). *Grid defense: N^d cost (exponential). Attack: $\leq 1/\delta$ (dimension-free). Ratio $\delta N^d \to \infty$.*

13 Experimental Validation

Validated against the Manifold of Failure framework [4] on three LLMs over a 2D behavioral space.

Llama-3-8B (AD 0.93, basin rate 93.9%): Tier 3 territory—boundary slope far exceeds $L(K+1)$ for reasonable K.

GPT-OSS-20B (AD 0.73, basin rate 64.3%): rugged landscape, many transversal crossings, each generating a persistent unsafe pocket.

GPT-5-Mini (peak AD 0.50, basin rate 0%): $U_{0.5} = \emptyset$, so no tier applies—correctly predicting no impossibility. Strategy: make the boundary harmless.

Table 1: Falsifiable predictions confirmed by empirical data.

Theorem	**Predicts**	**Confirmed by**
Basin (9.1)	Positive-measure basins	Extended heatmap regions
Fragment (9.2)	Smooth $\to$ large; rough $\to$ mosaic	Llama: mesa; GPT-OSS: mosaic
Convergence (10.2)	Monotone attack convergence	Curves plateau
Transfer (10.3)	Similar surfaces $\to$ shared basins	$.93 \to .73 \to .47$ spectrum
Authority (10.4)	Horizontal banding	Bands at $a_2 \approx .3, .7$
Persistent (6.2)	Steep boundary $\to$ unsafe volume	Llama's .93 plateau persists
Stability (11.1)	Deep points survive fine-tuning	Vulnerability persists across variants
Cost (12.1)	$d{=}2$ tractable; high-d not	15K queries fill 63% at $d{=}2$

14 Engineering Prescription

Make the boundary shallow (raise τ; GPT-5-Mini). **Reduce** L (smoother surface, predictable landscape). **Reduce dimension** (constrain prompt interface). **Monitor the boundary** (crossings persist under fine-tuning).

15 The Lean Artifact

Verified in Lean 4.28.0 + Mathlib v4.28.0. 36 files:

- 10 core (MoF_01–10), 10 cost (Cost_01–10), 10 advanced (Adv_01–10)
- MoF_ContinuousRelaxation (Tietze bridge)
- MoF_11_EpsilonRobust (ε-robust + persistent)
- MoF_12_Discrete (discrete path)
- 3 capstone (MasterTheorem, Euclidean, verification)

Three axioms: `propext`, `Classical.choice`, `Quot.sound`. Zero additional axioms.

16 Limitations

Tier 1 is at the boundary ($f(z) = \tau$ exactly).

Tier 2 constrains depth, not direction.

Tier 3 requires transversality ($c > L(K + 1)$).

Path B requires ordered structure; general graphs need further work.

Tietze applies to interpolants, not the discrete system itself.

Cost asymmetry assumes grid enumeration.

Single-turn. Manifold may shift across turns.

17 Conclusion

We have established defense impossibility via two independent paths.

Path A (continuous): boundary fixation $\to$ ε-robust band $\to$ persistent unsafe volume under transversality. Each tier strictly strengthens the last.

Path B (discrete): discrete IVT $\to$ defense fixation $\to$ capacity exhaustion. No topology, no Tietze, no continuity.

The key insight is not about attacks or training but about the *geometry of the defense itself*: design constraints create anchors that the defense cannot escape, whether the space is continuous or discrete.

The practical prescription: make the boundary shallow, smooth, and low-dimensional. GPT-5-Mini's ceiling at AD = 0.50 shows this working.

36 files, ∼300 theorems, zero `sorry`, three standard axioms.

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